

TEZPUR UNIVERSITY  
Test II, Autumn 2015  
C0 504 Natural Language Processing

*Marks : 25*

*Time : 30 minutes*

*The figures in the right-hand margin indicate marks for the individual question.*

1. What is the back-off technique in the context of smoothing of n-gram probabilities? Why may we combine back-off with, say, add-one smoothing?  
2 + 3
2. Suppose we decide to use variable code sizes for different symbols in a script using the concept of entropy. How can we do that? What will be the benefit?  
3 + 1
3. What are the broad word classes studied in traditional grammar of English? Point out further classification of two of these word classes.  
2 + 2

4. What are PoS tags? Write examples of typical tags. 1 + 1
5. How is PoS tagging done using rule-based approach? 2
6. Perform manual PoS tagging for the sentence- 3  
The small boys slowly wrote their names.
7. Give examples of a declarative sentence and an interrogative sentence in English, and write context free grammar rules that can recognise such sentences. 2 + 3